EXTERN + OUTAMATION
EXTERNSHIP

ADVANCED AI-POWERED

DOCUMENT INSIGHT AND DATA EXTRACTION PROJECT

SPRINT 7 STATUS UPDATE

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SKILLS

- Python
- AI/ML
- Data Scrapping
- Prompt Engineering
- LLMs
- Text Extraction
- Document Intelligence
- Python Automation
- Optical Character Recognition (OCR)
- Retrieval-Augumented Generation (RAG)

Helping businesses improve, automate, and architect critical tasks (such as document handling, customer service, and data management) by saving thousands of hours and boosting accuracy.

Mortgage Documents

SPRINTS

2025

01 AI Document Intelligence
02 Python & Google Colab
03 Python Data Extraction
04 Optimizing OCR
05 Implementing RAG
06 Optimizing RAG Pipelines
07 Blob Processing & Classification
08 Interactive Chatbot
09 AI-Powered Document Automation Platform

PROJECT TIMELINE

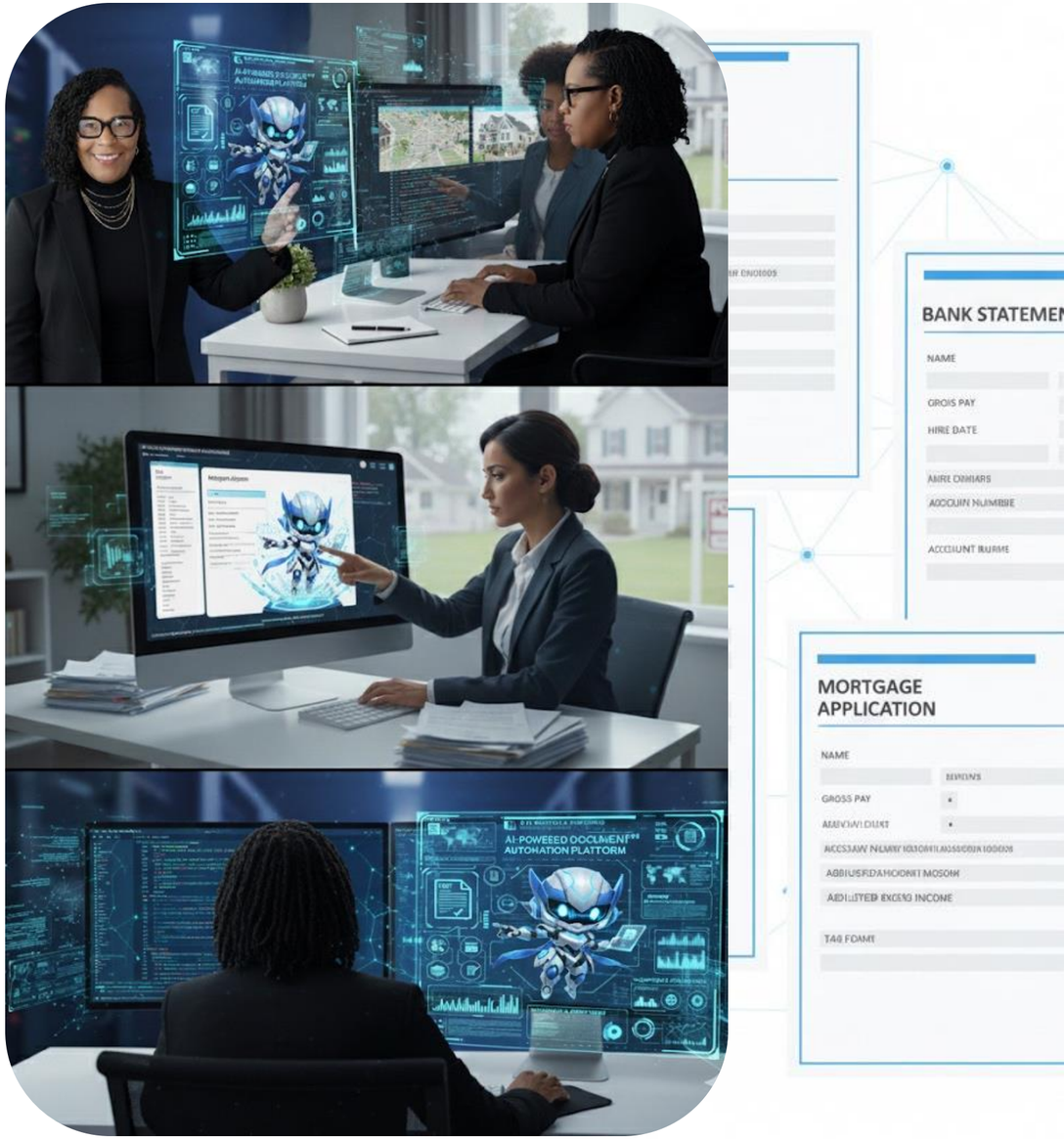
10 WEEKS

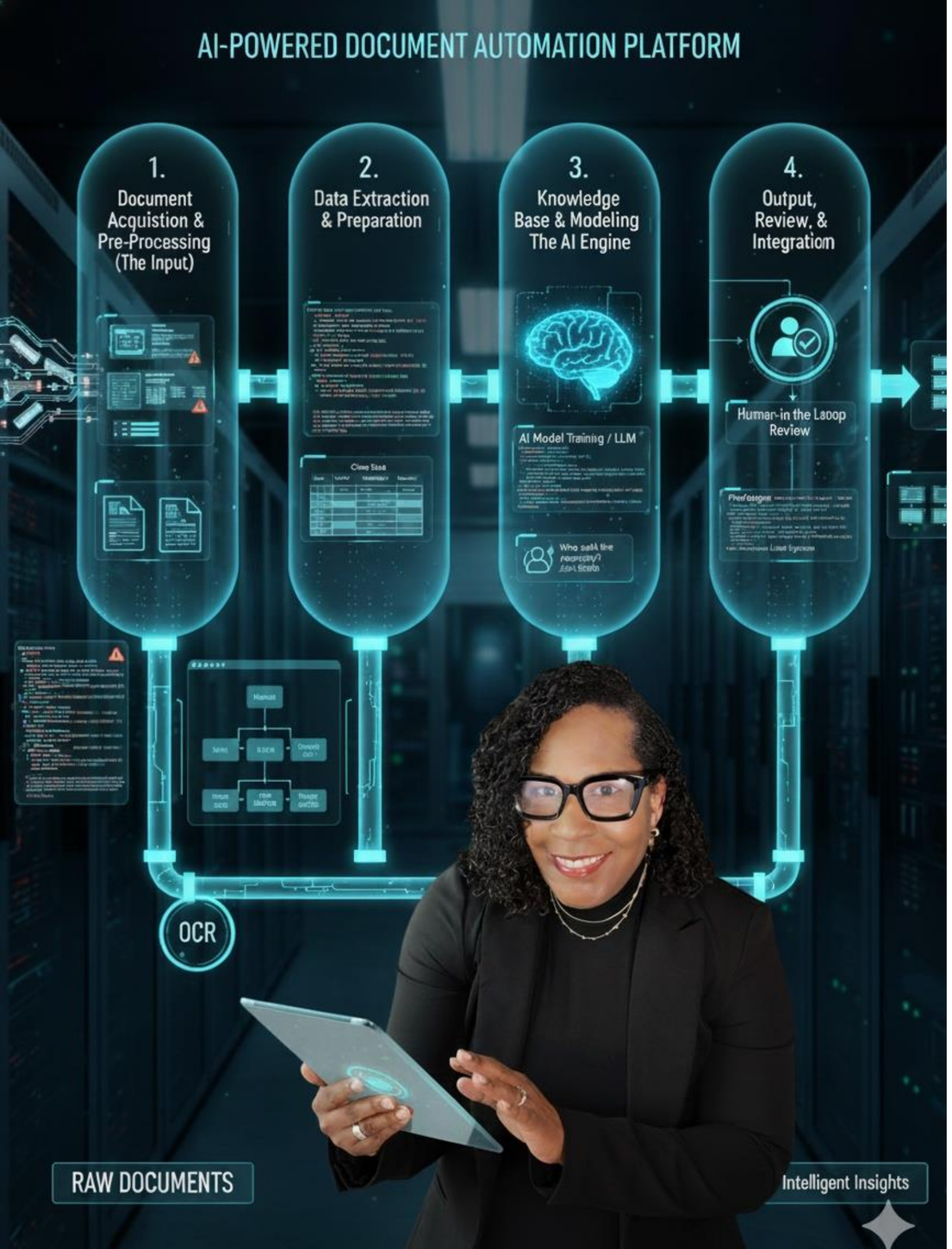
Blob Processing & Classification

Architecting Document Flow

Background: 10-Week Project

The Outamation Advanced AI-Powered Document Insights and Data Extraction Externship is a comprehensive 10-week program. Its core goal and purpose are to master the required skills to create an AI-Powered Document Automation Platform. The program focuses on bridging foundational technologies (ML, LLMs, NLP, Computer Vision) with high-value industry applications, such as solving complex, document-heavy challenges in mortgage automation, ultimately transforming slow, manual review processes into automated, efficient workflows.





Challenge & Solution: Building The Document Intelligence (AI) Pipeline

- **The Business Challenge:**
  - Automating the highly manual and time-consuming process of mortgage underwriting by handling 200-300 pages of unstructured documents.

- **The Solution:**
  - Designing and implementing a robust, AI-Powered Document Automation Platform to transform raw files into structured, intelligent insights.

- **The 4-Step Document Intelligence Workflow:**
  - Step 1. Document Acquisition & Pre-Processing (The Input)
  - Step 2. Data Extraction & Preparation
  - Step 3. Knowledge Base & Modeling (The AI Engine)
  - Step 4. Output, Review, & Integration (The Results)

SPRINT 7 FOCUS

Step 1 and Step 2 of Document Intelligence Workflow

The focus of Project 7 was to develop an intelligent system capable of segmenting unstructured, multi-document PDF "blobs" into individual files, classifying them by type, and implementing a metadata-driven routing pipeline for precise information retrieval.

Overall Status: Sprint 7 Complete

Phase 1: Split & Preprocess Multi-Document Blobs (Completed)

- Developed a system to identify document boundaries within massive, unstructured PDF "blobs."
- Implemented page-level metadata extraction using smart prompting.

Phase 2: Document Routing for PDF Files (Completed)

- Built a routing layer to send classified documents to specific extraction pipelines.
- Enabled targeted retrieval within large, complex mortgage files (e.g., loan packets, tax forms).

Final Output Status:

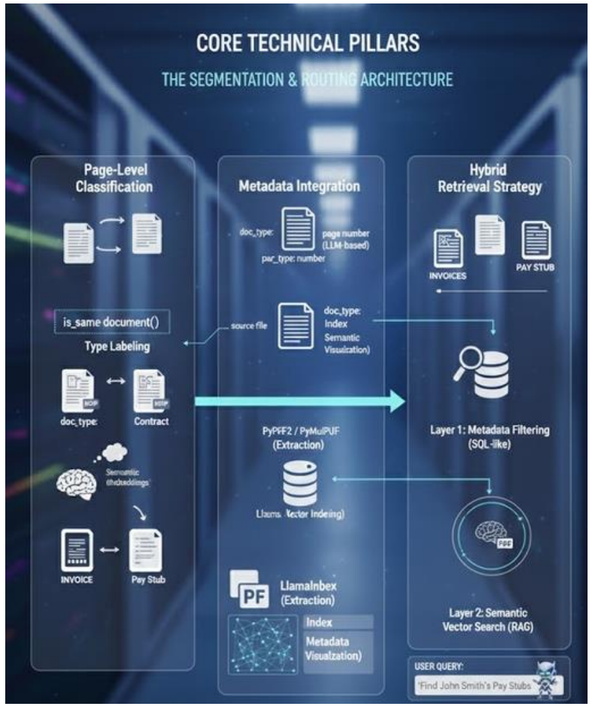
- Working RAG (Retrieval-Augmented Generation) pipeline for multi-document PDFs.
- Page-level index enriched with document-type metadata.

Key Highlights

Overcoming the "Blob" Problem

- **Structural Understanding:** Successfully transitioned from treating a PDF as one long text string to a structured collection of individual pages with distinct identities.
- **Enhanced Retrieval Precision:** By implementing page-level classification, the system avoids "haywire" retrieval and hallucinations caused by jumbled document types.
- **Metadata-Driven Routing:** Used metadata (doc_type, page_number) to filter queries, ensuring the LLM only "sees" relevant sections (e.g., searching only "Pay Stubs" for income questions).
- **Efficiency & Scalability:** Developed a "search where it matters" strategy that is faster and more cost-effective than embedding every page of every document.

Core Technical Pillars

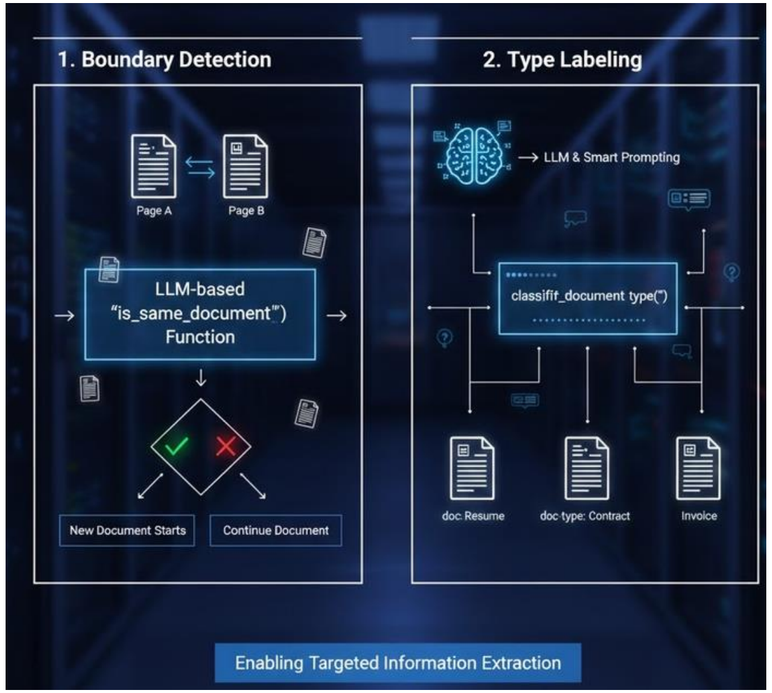


The Core Technical Pillars section outlines the foundational engineering that enables the transition from unstructured PDF "blobs" to a highly organized, searchable intelligence asset. This framework is built upon a multi-layered approach that integrates Large Language Model (LLM) reasoning with precise metadata tagging. By anchoring the project on advanced segmentation logic and a hybrid retrieval strategy, we ensure maximum accuracy and operational speed when navigating the complexities of high-stakes mortgage and legal documentation.

The Segmentation & Routing Architecture

- Page-Level Classification
- Metadata Integration
- Hybrid Retrieval Strategy

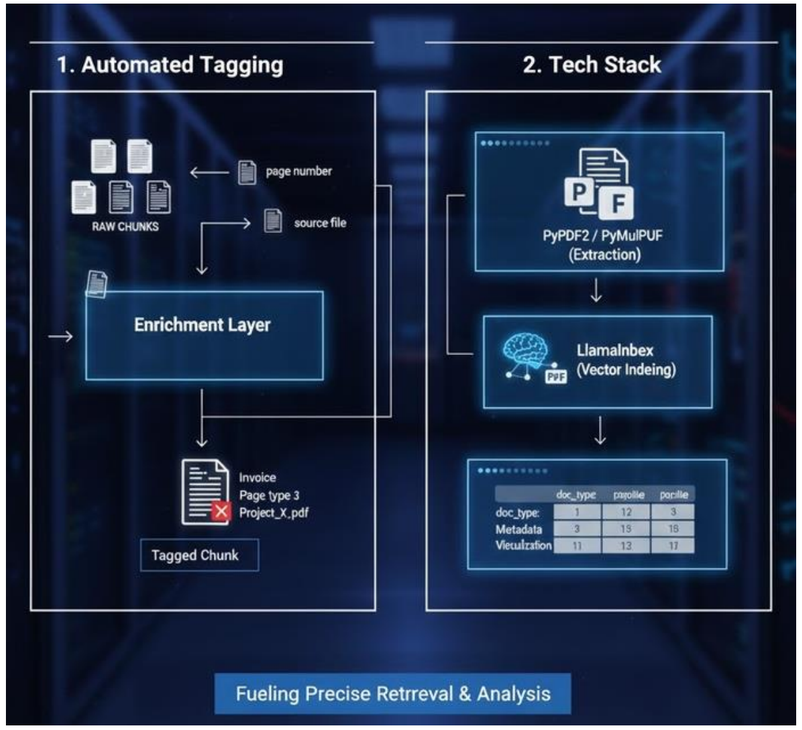
Page-level Classification



- **Boundary Detection:** An LLM-based is_same_document() function compares consecutive pages to determine if a new document starts.

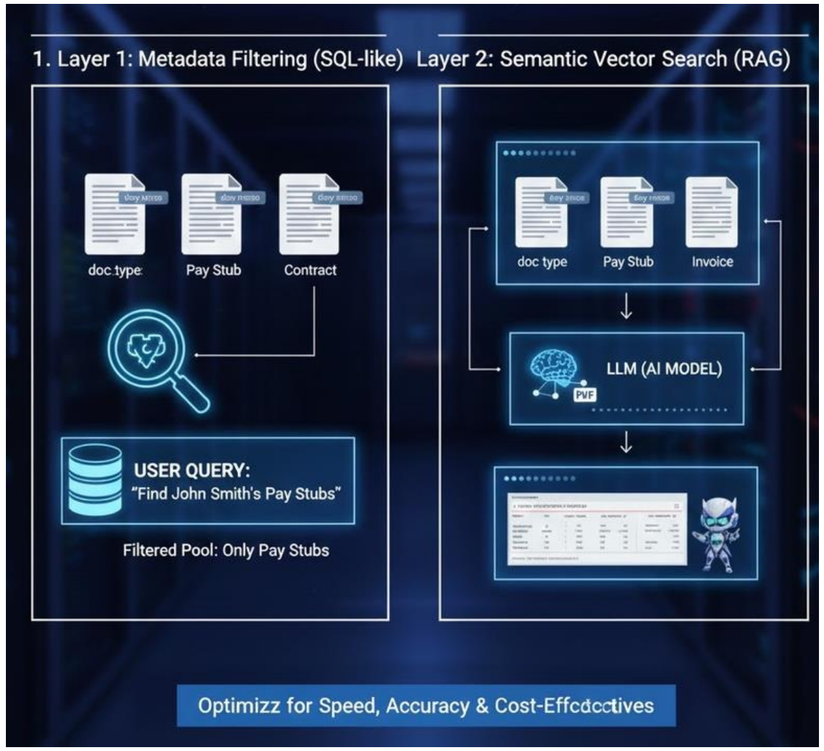
- **Type Labeling:** A classify_document_type() function assigns labels like "Resume," "Contract," or "Lender Fee Sheet."

Metadata Integration



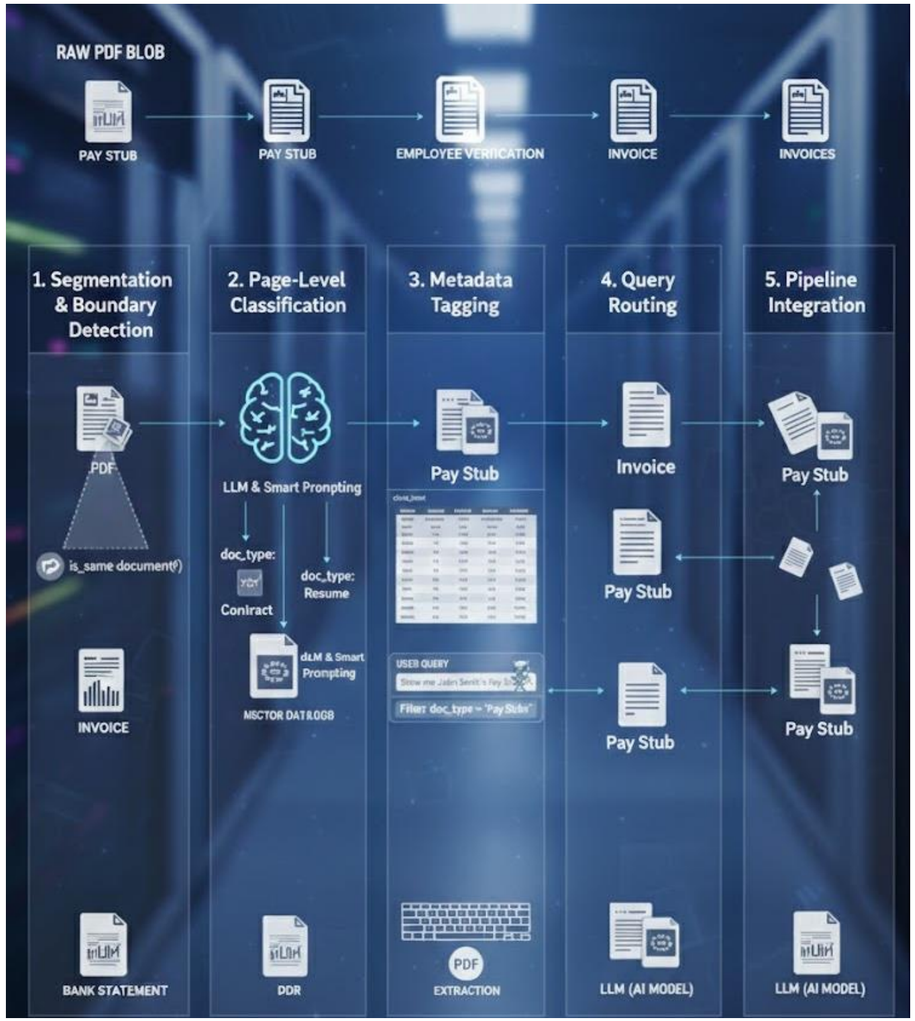
- **Each Chunk:** tagged with doc_type, page_number, and source_file before indexing.
- **Tools Used:** PyPDF2 or PyMuPDF for extraction, LlamaIndex for vector indexing, and pandas for metadata visualization.

Hybrid Retrieval Strategy



- **First Layer:** Fast metadata filtering (e.g., SQL-like filtering) to narrow down the document pool.
- **Second Layer:** Semantic vector search (RAG) applied only to the filtered results to handle vague user queries.

Key Tasks

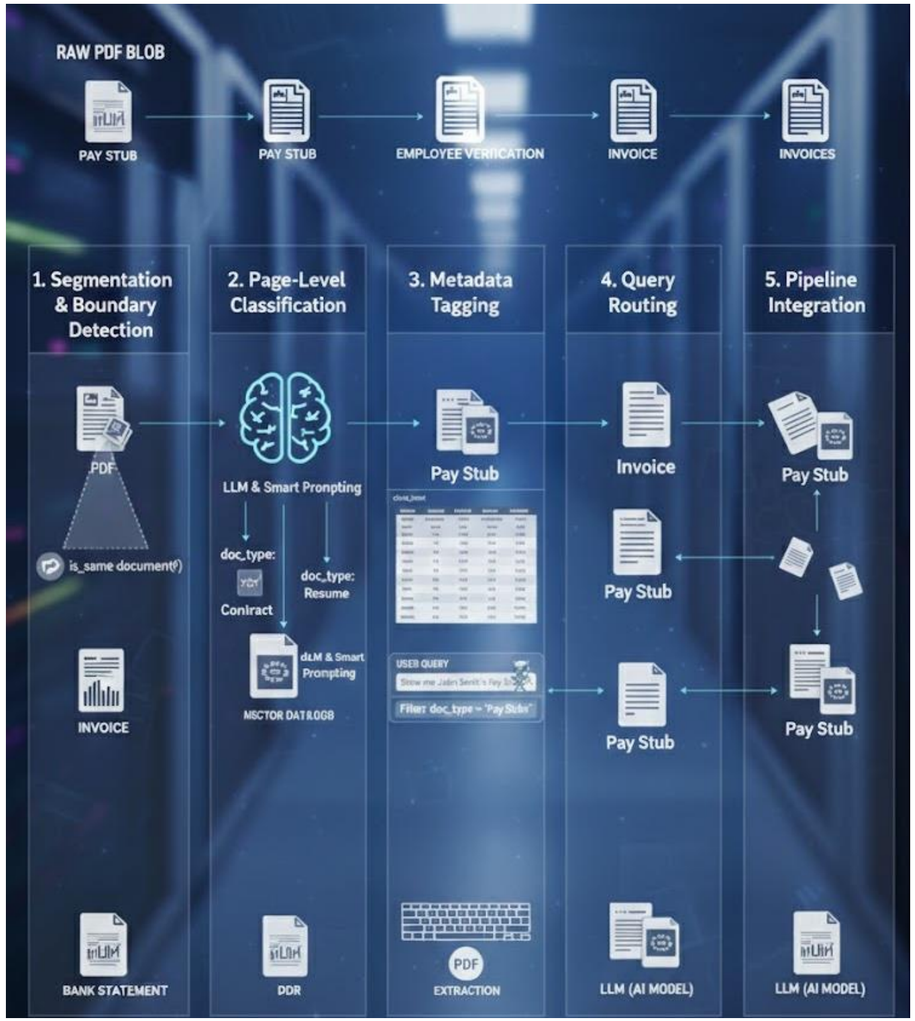


- **Document Segmentation and Boundary Detection:** Implementing a strategy to analyze PDFs page-by-page to determine where one document ends and another begins.

- **Page-Level Classification:** Using Large Language Models (LLMs) and smart prompting to identify the specific type of each document (e.g., Resumes, Contracts, W-2s, or Pay Stubs).

- **Metadata Tagging:** Enriching document chunks with searchable metadata such as doc_type, page_number, and source_file to enable filtered retrieval.

Key Tasks



- **Query Routing:** Building a function that classifies a user's query and routes it only to the relevant document types, reducing hallucinations and processing costs.

- **Pipeline Integration:** Combining these elements into a full Retrieval-Augmented Generation (RAG) system that handles complex, mixed-document files.

Core Deliverables

- **Working RAG Pipeline:** A functional system that can ingest messy, multi-document PDFs and provide accurate, targeted answers.
- **Enriched Page-Level Index:** A vector database index where every chunk is correctly labeled with its document type and location.
- **Metadata Mapping:** A structured output (JSON or pandas DataFrame) that provides a clear map of every page's identity within the original blob file.
- **Technical Documentation:** A summary report detailing the segmentation strategy, model selection (such as open-source models like Mistral or Llama), and logic for the retrieval system.

CoLab Notebook

- **Page-Level Detection Strategy Using RAG**
  - https://github.com/LashawnFofung/RAG-Pipelines/blob/main/src/Task_Designing_a_Page_Level_Detection_Strategy_Using_RAG.ipynb

- **Tagging Chunks with Metadata in LlamaIndex**
  - https://github.com/LashawnFofung/RAG-Pipelines/blob/main/src/Task_Tagging_Chunks_with_Metadata_in_LlamaIndex.ipynb

- **Route Queries**
  - https://github.com/LashawnFofung/RAG-Pipelines/blob/main/src/Task_Route_Queries.ipynb

- **End-to-End RAG Pipeline with Page-Level Metadata and Query Routing**
  - https://github.com/LashawnFofung/RAG-Pipelines/blob/main/src/Task_End_to_End_RAG_Pipeline_with_Page_Level.ipynb

THANK YOU

https://github.com/LashawnFofung/AI-Powered-Document-Automation-Platform